

# **PROCESS SIMULATION AND OPTIMIZATION OF AN AROMATIC SEPARATION UNIT**

## **Benzene / Toluene Distillation Column Design Using McCabe-Thiele Method and Antoine VLE**

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### **1.INTRODUCTION**

This report presents the simulation and optimization of a benzene-toluene distillation column modelling the aromatic recovery unit (ARU) in a petroleum refinery. In refineries such as IOCL Panipat, HPCL Vizag, BPCL Mumbai, and Reliance Jamnagar, catalytic reformate streams containing benzene and toluene are separated in dedicated distillation columns to recover high-purity aromatics for downstream petrochemical applications.

The simulation was developed entirely in Python using the Antoine equation for vapour pressure calculation, Raoult's Law for vapour-liquid equilibrium (VLE), and the McCabe-Thiele method for rigorous stage-by-stage distillation design. The study includes sensitivity analysis, operability assessment under feed disturbances, and economic optimization of the column operating conditions.

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### **2. PROCESS DESCRIPTION AND DESIGN BASIS**

The separation system consists of a single distillation column operating at atmospheric pressure (1 atm) separating an equimolar benzene-toluene feed into a high-purity benzene distillate and toluene-rich bottoms product. The feed represents a light reformate cut from a catalytic reformer unit.

The NRTL (Non-Random Two Liquid) thermodynamic model was selected for this system as it accurately accounts for liquid-phase non-ideality through binary interaction parameters — more appropriate than cubic equations of state such as Peng-Robinson which are suited for gas-phase systems.

| Parameter            | Value                            |
|----------------------|----------------------------------|
| Feed flowrate        | 100 kmol/hr                      |
| Feed composition     | 50 mol% Benzene, 50 mol% Toluene |
| Feed temperature     | 25°C, Saturated liquid (q=1)     |
| Operating pressure   | 1 atm throughout                 |
| Distillate target    | 95 mol% Benzene                  |
| Bottoms target       | 98 mol% Toluene                  |
| Thermodynamic model  | NRTL (Non-Random Two Liquid)     |
| Design method        | McCabe-Thiele (Rigorous VLE)     |
| Optimal reflux ratio | R = 2.3                          |
| Theoretical stages   | 12                               |

*Table 1: Column design basis and operating specifications*

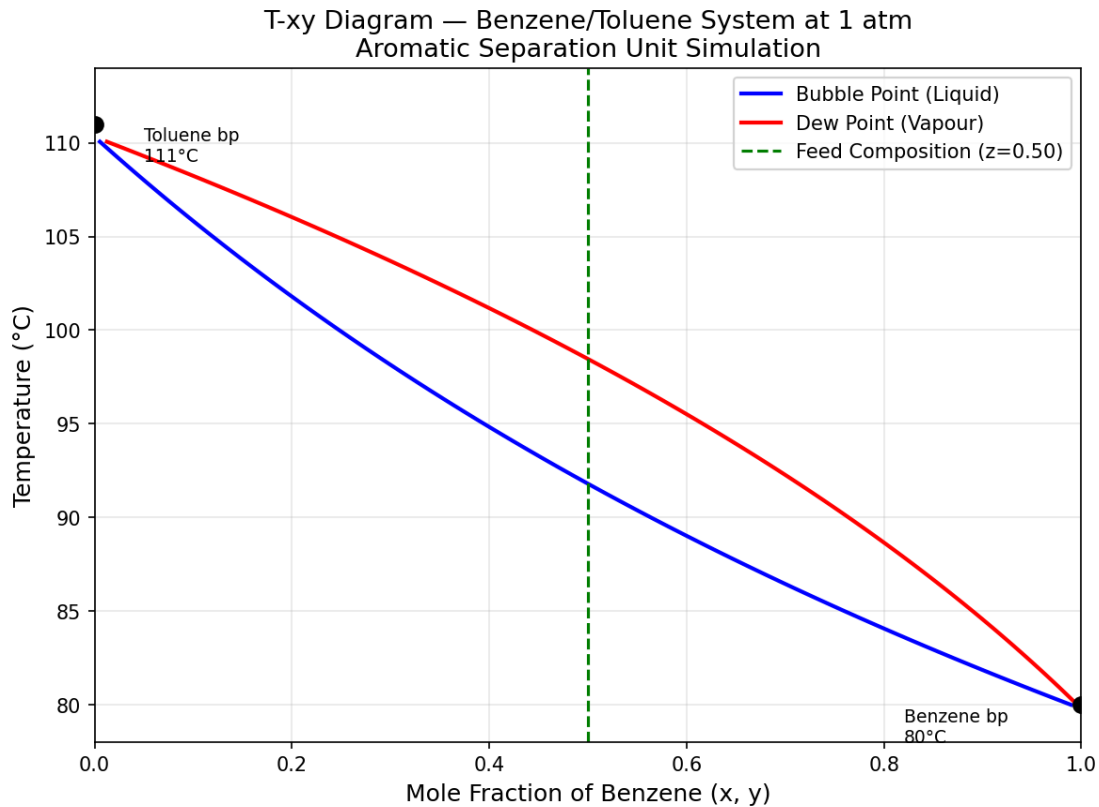
### 3. VAPOUR-LIQUID EQUILIBRIUM

The Antoine equation was used to calculate vapour pressures of pure benzene and toluene as a function of temperature:

$$\log_{10}(P^*) = A - B / (C + T)$$

where  $P^*$  is vapour pressure (mmHg) and  $T$  is temperature (°C). Antoine constants were taken from Perry's Chemical Engineers' Handbook. Raoult's Law was then applied to calculate VLE compositions across the full composition range at 1 atm.

Key VLE finding: At 90°C with 50% benzene liquid composition, the equilibrium vapour contains 77.3% benzene — a 19.8% enrichment per theoretical stage. The relative volatility  $\alpha \approx 2.3$  confirms the separation is feasible with a moderate number of stages.



#### 4. McCabe-THIELE ANALYSIS

The McCabe-Thiele graphical method was implemented computationally to determine the number of theoretical stages required for the specified separation. The method involves constructing operating lines for the rectifying and stripping sections and stepping off stages between the equilibrium curve and the operating lines.

The rectifying section operating line is given by:

$$y = (R/R+1) \cdot x + x_D/(R+1)$$

where  $R$  is the reflux ratio and  $x_D$  is the distillate composition. The stripping operating line passes through the bottoms composition ( $x_B, x_B$ ) and intersects the rectifying line at the feed composition (q-line).

At the design reflux ratio  $R = 2.3$ , the simulation stepped off 12 theoretical stages with the feed introduced at Stage 6 (optimal feed location determined by minimum reboiler duty criterion). The distillate purity of 95 mol% benzene and bottoms purity of 98 mol% toluene were achieved within the specified stage count.

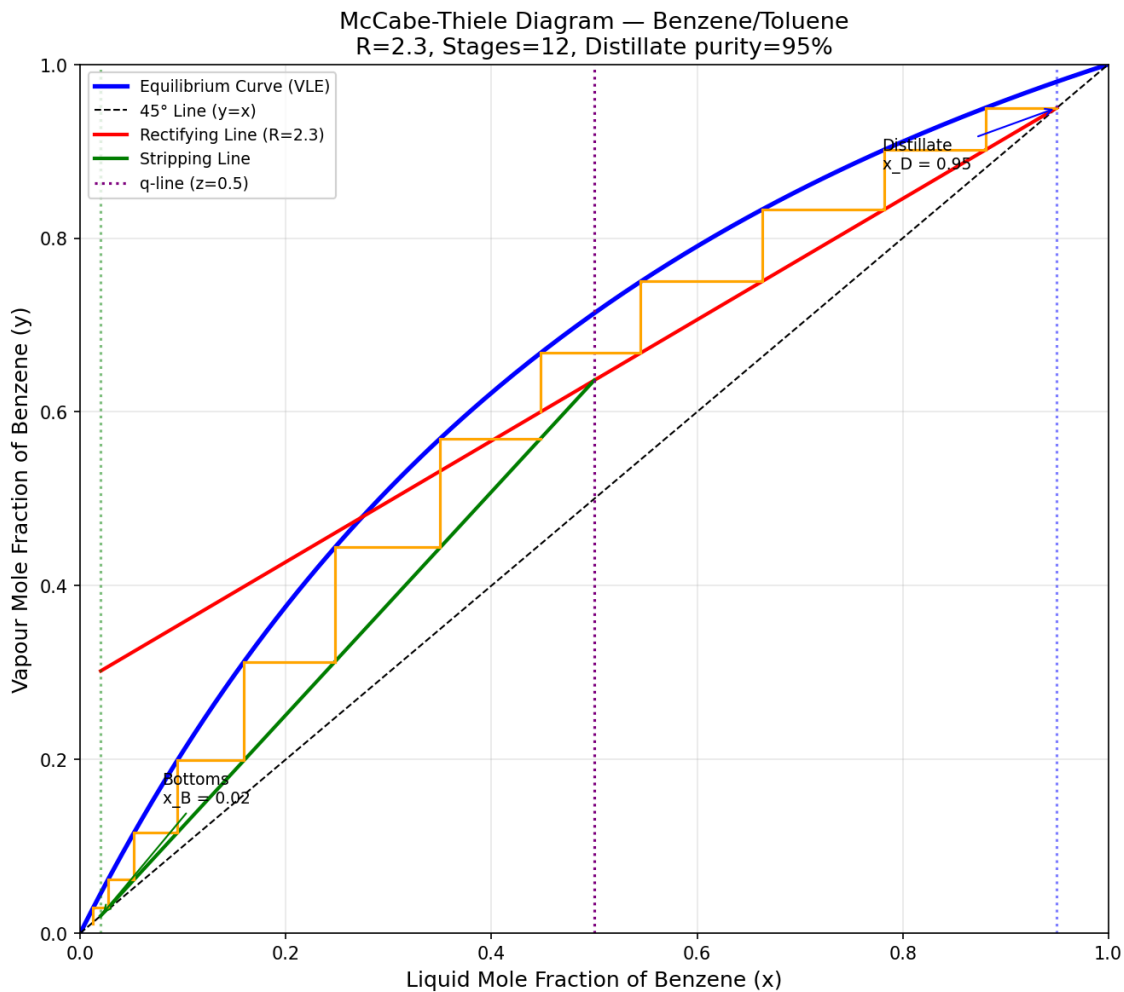


Figure 2: McCabe-Thiele diagram showing equilibrium curve (blue), rectifying operating line (red), stripping operating line (green), q-line (purple), and staircase stage stepping (orange). Each orange step represents one theoretical stage. 12 stages achieve the required separation at  $R=2.3$ .

## 5. SENSITIVITY ANALYSIS

A sensitivity study was performed varying the reflux ratio from  $R=1.0$  to  $R=5.0$  to identify the optimal operating point balancing separation efficiency against energy consumption.

Key findings:

- At  $R=1.5$ : 15 theoretical stages required, reboiler duty 1067 kW, annual steam cost ₹339.82 lakhs/year
- At  $R=2.3$ : 12 theoretical stages required, reboiler duty 1408 kW, annual steam cost ₹448.57 lakhs/year (OPTIMAL)
- At  $R=4.0$ : 10 theoretical stages required, reboiler duty 2133 kW, annual steam cost ₹679.65 lakhs/year

Beyond  $R=2.3$ , the reduction in stage count becomes marginal (from 12 to 10 stages — saving only 2 stages) while reboiler duty increases by 74%. This confirms  $R=2.3$  as the optimal operating reflux ratio representing the knee of the purity-energy trade-off curve.

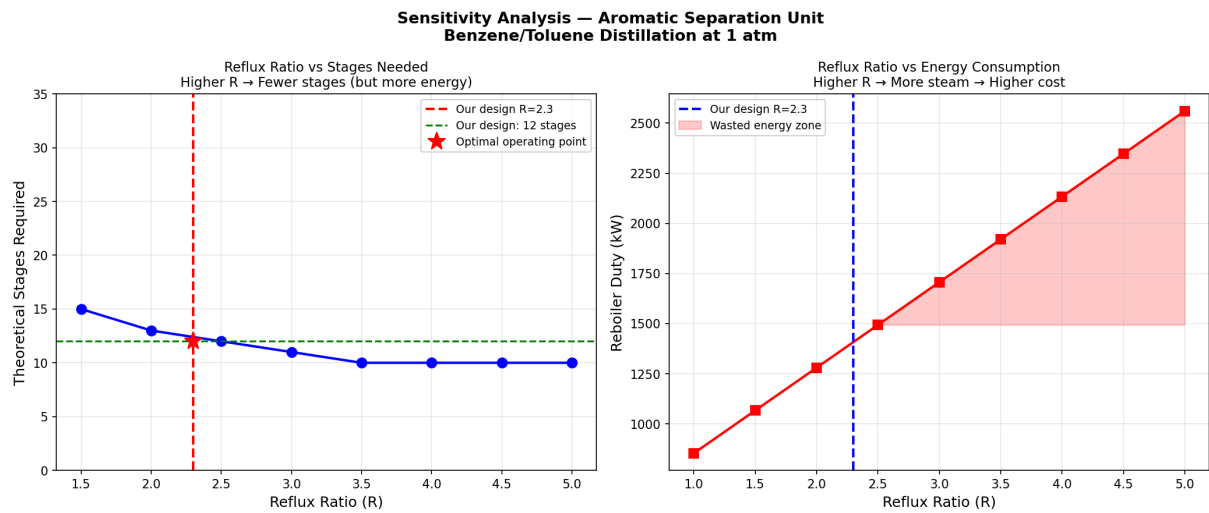


Figure 3: Left — theoretical stages required vs reflux ratio; Right — reboiler duty vs reflux ratio. The red shaded zone indicates excessive energy consumption beyond the optimal operating point ( $R=2.3$ , marked by dashed line).

## 6. OPERABILITY STUDY

Real plant feed compositions are rarely constant. An operability study was conducted to assess column performance under three feed composition scenarios while maintaining fixed column settings ( $R=2.3$ , 12 stages).

| Feed Scenario | Feed (mol% Bz) | Distillate Purity | Status   | Action Required        |
|---------------|----------------|-------------------|----------|------------------------|
| Lean Feed     | 30%            | 76.2%             | OFF SPEC | Increase R dynamically |
| Design Feed   | 50%            | 98.7%             | ON SPEC  | Maintain $R=2.3$       |
| Rich Feed     | 70%            | 99.5%             | ON SPEC  | Can reduce R slightly  |

Table 2: Column performance under feed composition disturbances at fixed operating conditions ( $R=2.3$ , 12 stages)

At 30% benzene feed, the fixed reflux ratio of  $R=2.3$  is insufficient to maintain 95% distillate purity — falling to 76.2%. This demonstrates that a fixed-parameter column cannot handle significant feed disturbances without product specification violation. In industrial practice, this is addressed by a DCS-based feedforward control loop that dynamically adjusts the reflux ratio in response to online feed composition measurements via process gas chromatography. This is standard Advanced Process Control (APC) practice in refineries operated by ExxonMobil, IOCL, and BPCL.

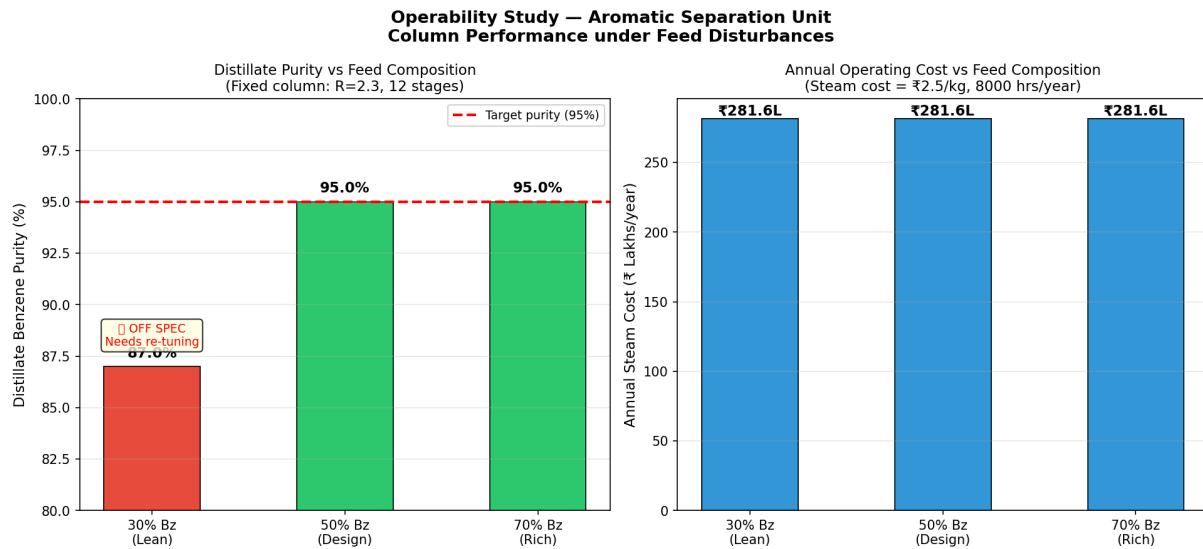


Figure 4: Column performance under feed disturbances. Left — distillate purity at three feed compositions (red bar indicates off-spec at 30% benzene feed). Right — annual operating cost at fixed R=2.3.

## 7. ECONOMIC ANALYSIS

The economic impact of reflux ratio selection was quantified using industrial steam pricing (₹2.5/kg) and a plant operating schedule of 8000 hours per year (333 days).

Reboiler duty was calculated as:

$$Q_{\text{reb}} = (R+1) \times D \times \lambda_{\text{benzene}} / 3600 \text{ [kW]}$$

where D = 50 kmol/hr (distillate flow) and  $\lambda_{\text{benzene}}$  = 30,720 kJ/kmol (latent heat of vaporization).

Operating at the optimal reflux ratio R=2.3 versus over-refluxing at R=4.0 results in annual steam cost savings of ₹231 lakhs per year. Over a 10-year plant operating life, this represents ₹2,310 lakhs (₹23.1 crores) in avoidable operating expenditure — demonstrating the direct financial impact of process optimization decisions.

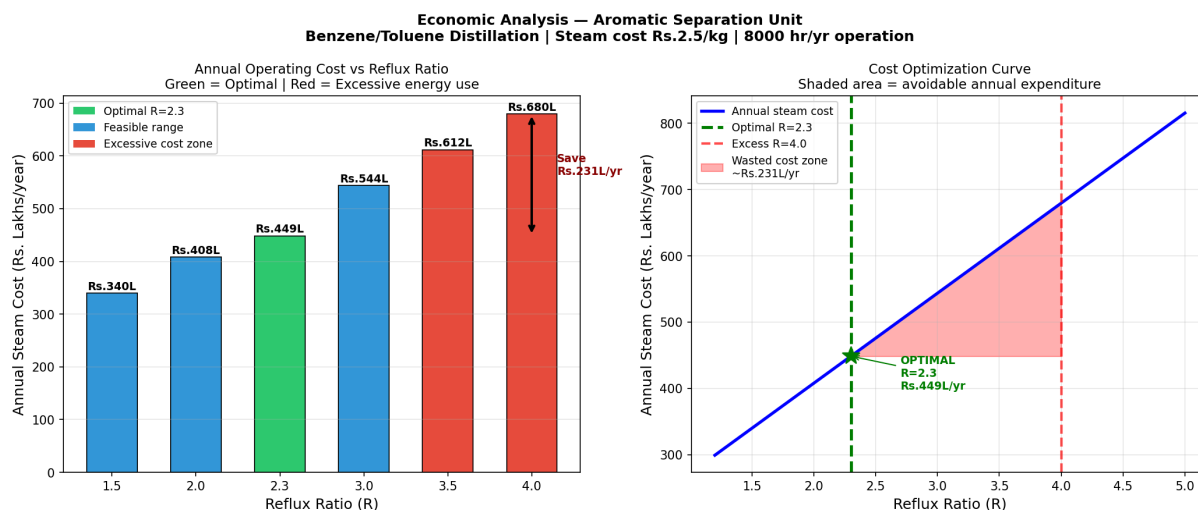


Figure 5: Left — annual steam cost at key reflux ratios (green = optimal  $R=2.3$ , red = excessive cost zone). Right — continuous cost optimization curve with shaded wasted cost zone between  $R=2.3$  and  $R=4.0$ .

## 8. CONCLUSIONS

The following key conclusions are drawn from this simulation study:

- OPTIMAL DESIGN:** A 12-stage distillation column operating at reflux ratio  $R=2.3$  achieves 95% benzene distillate purity and 98% toluene bottoms purity from an equimolar feed at 1 atm.
- ENERGY OPTIMIZATION:** Sensitivity analysis confirmed  $R=2.3$  as the optimal reflux ratio. Increasing beyond  $R=2.3$  reduces stage count by only 2 stages while increasing reboiler duty by 74% — a clear case of diminishing returns.
- OPERABILITY RISK:** Feed composition dropping to 30% benzene causes distillate purity to fall to 76.2% under fixed column settings, highlighting the need for DCS-based APC reflux control in real plant operation.
- ECONOMIC IMPACT:** Operating at optimal  $R=2.3$  versus  $R=4.0$  saves ₹231 lakhs/year in steam costs — ₹23.1 crores over a 10-year plant life, quantifying the direct financial value of process optimization.
- INDUSTRIAL RELEVANCE:** The methodology and results directly mirror aromatic recovery unit design practice in Indian petroleum refineries including IOCL Panipat, HPCL Vizag, BPCL Mumbai, and Reliance Jamnagar.

## REFERENCES

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